Dr. Soly Mathew - Curriculum Vitae

Materials Scientist | Flexible Sensor Researcher
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ACADEMIC & RESEARCH APPOINTMENTS

Scientific Officer - Indian Institute of Heritage, Delhi (Jul 2026 - Present)
Dept of Art Conservation: Scientific examination, characterization & degradation analysis

Project Associate - Indian Institute of Technology Jodhpur (IIT Jodhpur) (Oct 2025 - Jun 2026)
R&D on functional nanocomposites, sensor design & electromechanical characterisation

Assistant Professor (Adhoc) - NSS College of Engineering, Palakkad (Aug 2025 - Oct 2025)
Undergraduate Engineering Physics curriculum delivery, laboratory instruction & mentoring

EDUCATION

PhD in Materials Science - Vellore Institute of Technology (VIT), India (2020 - Present)
Focus: Flexible capacitive pressure sensors, polymer nanocomposites, ternary dielectrics

PEER-REVIEWED JOURNAL PUBLICATIONS

1. Smart Materials and Structures (2026) - DOI: 10.1088/1361-665X/ae8228
2. Journal of Applied Polymer Science (2026) - DOI: 10.1002/app.70439
3. Flexible and Printed Electronics (2025) - DOI: 10.1088/2058-8585/adc81e
4. Sensors International (2025) - DOI: 10.1016/j.sintl.2025.100330
5. IEEE Access (2025) - DOI: 10.1109/ACCESS.2025.3577066

CORE SCIENTIFIC & DIGITAL RESEARCH COMPETENCIES

Materials: Polymer Nanocomposites, PDMS, PVDF, ZnO, TiO2, ZrO2, Graphene Oxide, Carbon Black
Fabrication: Spin Coating, Solution Casting, Cleanroom Thermal Curing, Vacuum Degassing
Characterisation: FESEM, XRD, FTIR, BDS Broadband Dielectric Spectroscopy, LCR Testing
Scientific Computing: Python 3, JupyterLab, NumPy, Pandas, SciPy, Matplotlib, Claude Code, Copilot

TARGET INSTITUTIONAL COLLABORATION

Danish & European Research Institutes: DTI, FORCE Technology, Coloplast, DTU, SDU, AU